

Nadeem Mohamed Soliman

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PROFILE

Software Engineer and Computer Science and Engineering graduate from the German University in Cairo, with an **A+ Bachelor's degree**. I have shipped full-stack applications, backend microservices, data pipelines, and AI/ML systems across banking, logistics, and enterprise software, working across the complete software development life cycle (SDLC) in Agile/Scrum teams. My work spans REST and GraphQL API design, distributed systems, cloud deployment (GCP, AWS), and applied machine learning – from production LLM pipelines to computer vision research. I write unit and integration tests as a habit, not an afterthought, and care about code the next engineer can actually read.

EDUCATION

German University in Cairo (GUC)

Cairo, Egypt

Bachelor of Science in Computer Engineering and Science

Sep 2021 – Jun 2026

- **Cumulative GPA: 1.24 / 5.0** (Top of class; German grading scale where 1.0 is the highest possible grade)
- Bachelor Thesis: Grade **A+** in Vision Transformers for Autonomous Vehicle Intention Prediction
- Relevant Coursework: Cloud Computing, Data Structures and Algorithms, Operating Systems, Computer Networks, Database Systems, Network Security, Software Engineering, Artificial Intelligence, Project Management

BACHELOR THESIS

Vision Transformer Framework for Vehicle Intention Prediction

PyTorch, Python, Computer Vision, Deep Learning

Feb 2025 – Jun 2025

- Reproduced Uber's CNN baseline from the original paper, retrained it on the public Argoverse 2 autonomous driving dataset, and trained a Vision Transformer (ViT) under identical conditions to benchmark both architectures on vehicle maneuver/intention prediction.
- Designed and implemented the full data preprocessing, temporal feature extraction, model training, and evaluation pipeline for both architectures; at matched training checkpoints, the ViT outperformed the CNN baseline by **52.9%**.
- Conducted comparative model benchmarking and error analysis to validate results against published research, applying rigorous experiment tracking and reproducibility practices.

EXPERIENCE

Executive Software Engineer | Aman Raya

June 2026 – Present

- Developing **Spring Boot** microservices for the Bill Payments team, integrating external billers and service providers through **WSO2** to enable secure bill payment processing backed by **PostgreSQL**.

Software Engineer, Part Time | Turuq

June 2026 – Present

- Refactored and consolidated WhatsApp, SMS, and email channels into a single modular **TypeScript** backend service using inheritance and polymorphism, enabling new channels to be added without modifying existing integrations, reducing code duplication, and improving maintainability.
- Built a 54-test black-box integration test suite using **Vitest** and **Supertest**, exercising the communication service over HTTP while validating MongoDB side effects and outbound Graph API calls, enabling reliable regression testing and CI integration.

Software Engineer | Turuq

Oct 2025 – Jun 2026

- Built an embedded Shopify app using **Remix, React, and Node.js** that connects merchant storefronts to a logistics backend via Shopify Webhooks and the GraphQL Admin API, processing real-time order events across multiple merchant accounts.
- Designed and shipped a bulk waybill generation feature that automated the end-to-end shipment creation workflow, reducing manual processing time by **80%**.
- Developed backend features for a courier management service built with **Bun, Hono, Clerk, MongoDB, and Next.js**, supporting real-time order tracking and courier dispatch.
- Built a customer communication service on top of the **Meta Cloud API** for inbound/outbound messaging, stateful conversation workflows, and HMAC-verified webhooks, enabling automated customer notifications and order confirmations.

Backend Developer Intern | Ejada

Aug 2025

- Built scalable **Spring Boot** microservices for a virtual banking platform, integrating **Apache Kafka** for asynchronous event-driven messaging and **WSO2 API Gateway** for secure external API routing.
- Developed the **REST API** layer for mobile and web banking clients, implementing **OAuth 2.0** authentication, role-based access control (RBAC), and request validation following secure software development lifecycle (SSDLC) practices.

Backend Developer Intern | Link Development

Jul 2025 – Aug 2025

- Integrated **Keycloak** single sign-on (SSO) into **OpenKM**, centralizing identity and access management within a Dockerized MSSQL environment.
- Built automated backup and disaster recovery pipelines using **Rclone** to minimize data-loss risk and support compliance requirements.
- Authored deployment and onboarding documentation that became the team's standard reference, reducing new-engineer ramp-up time by **60%**.

System Development Intern | Electrolux

Aug 2024 – Sep 2024

- Designed and built an automated quality tracking system in **VBA (Excel and Access)**, replacing a manual process and reducing data entry time by **70%** while improving reporting accuracy.

Full Stack Intern | QNB Egypt

Jul 2024

- Developed a customer survey platform using **Java**, **XHTML**, and **MySQL** that enabled branch teams to collect, store, and review structured customer feedback.

Technical Intern | BVS Company

Jul 2024 – Sep 2024

- Gained hands-on experience with **IBM DataPower API Gateway**, ACH payment processing workflows, and enterprise banking security protocols.

Software Development Intern | The Sparks Foundation

Jul 2023

- Developed a full-stack **MERN** banking application featuring user authentication, account management, and transaction history, delivering the project within a one-month sprint.

UNIVERSITY PROJECTS**Massively Scalable Food Delivery Platform (Talabat-like) | *Spring Boot, PostgreSQL, MongoDB, Redis, RabbitMQ, Feign***

- Contributed to a distributed, fault-tolerant food delivery backend built as independently deployable microservices for customers, restaurants, and drivers; owned the **User Service** end to end – JWT authentication, BCrypt password hashing, and role-based access control.
- Replaced shared-table SQL joins with **OpenFeign** clients for inter-service communication, published domain events over **RabbitMQ**, and wrote idempotent Saga-pattern consumers to update user stats on order completion and cancellation, backed by a dead-letter queue (DLQ) with three-attempt retry for fault tolerance.
- Deployed the User Service on **Kubernetes** with health/readiness probes and a headless StatefulSet for Postgres; added **Prometheus** metrics with a **Grafana** dashboard and structured **Loki** logs with MDC fields for full observability.

Agentic NDA Contract Analysis System | *LLMs, QLoRA, LangChain, Agentic AI, NLP, Vector Search*

- Contributed to a 5-member team building an agentic contract review pipeline with reasoning, policy, orchestration, and logging components – an applied **retrieval-augmented generation (RAG)** and **agentic workflow** system for legal document analysis.
- Fine-tuned a Qwen large language model (LLM) using **QLoRA** parameter-efficient fine-tuning on the ContractNLI benchmark to classify 17 legal hypotheses per contract as ENTAILED, CONTRADICTED, or NOT MENTIONED.
- Built the vector embedding and semantic search layer for retrieval over contract clauses using a vector database, focusing on output reliability and grounding across real-world legal documents.

Real-Time Stock Analytics Platform | *Kafka, PySpark, Airflow, Docker, LangChain, PostgreSQL*

- Contributed to an end-to-end **data engineering** pipeline using **Apache Airflow** for workflow orchestration and **Apache Kafka** with **PySpark** for real-time market data stream processing.
- Stored processed data in **PostgreSQL** and helped integrate a **LangChain** Text-to-SQL agent for natural language querying over the warehouse; fully containerized with **Docker** and cloud-deployed.
- Built interactive **Apache Superset** dashboards to visualize streaming market data and pipeline health metrics for data-driven monitoring.

Graph-RAG Travel Assistant | *Python, Neo4j, LangChain, Explainable AI*

- Contributed to a **knowledge graph** built from 50,000+ hotel reviews with named entity recognition (NER) and a **Streamlit** frontend for interactive querying.
- Helped design a hybrid retrieval layer combining vector similarity search with Cypher-based graph traversal to ground LLM answers in structured data, reducing hallucination.
- Added **SHAP** and **LIME** explainability (XAI) to the underlying ML models to make recommendations interpretable and auditable.

All-in-One Vacation Platform | *MERN, Stripe API, Amadeus API*

- Contributed to a full-stack travel booking platform where users can search and book flights and hotels, apply promo codes, and receive personalized recommendations.
- Integrated **Stripe** for payment processing and **Amadeus** for live flight and hotel booking data via third-party API integration.
- Worked in an Agile team using Git version control and GitHub for collaboration, contributing across the MERN stack from API design to frontend components.

Embedded Auto Lane Detection | *C, Computer Vision, Embedded Systems, Hardware Integration*

- Contributed to a real-time lane detection system written in **C** that interfaced directly with hardware drivers on an embedded platform, working within tight memory and real-time processing constraints.
- Helped implement the image processing and signal processing logic to identify lane boundaries and feed steering signals to the navigation layer.
- Tested the system under varying lighting and road conditions to validate detection reliability and robustness.

University Academic System and Portal | *SQL Server, JavaScript, HTML, CSS*

- Contributed to an academic management system with a normalized relational database schema and **50+ stored procedures**, views, triggers, and functions.
- Automated registration, graduation checks, and academic probation rules at the database layer to enforce business logic consistently across the application.
- Helped build a role-based web portal with separate views and permissions for students, advisors, and administrators.

Donation Ecosystem Platform | *React.js, Accessibility (WCAG)*

- Contributed to a platform connecting donors with NGOs, taking part in requirements gathering across stakeholder groups and overall system design.
- Helped deliver a responsive, accessible **React** prototype covering donor registration, campaign browsing, and NGO profiles, following web accessibility (WCAG) standards.
- Coordinated with teammates to align reusable, component-based UI architecture with the agreed data model and user flows.

The Last of Us: Legacy Game Engine | *Java, OOP, MVC Architecture*

- Contributed to a turn-based survival game in **Java** applying object-oriented programming (OOP) principles and **MVC** software architecture.
- Helped build the core game loop, grid-based map, and entity interaction system covering players, enemies, and collectibles.
- Implemented a custom GUI and thorough exception handling to keep game state consistent across edge cases.

VOLUNTEERING AND LEADERSHIP

Director, Juniors Technical Program | IEEE GUC Student Branch

Sep 2024 – Jul 2025

- Led and managed a **40-person** cross-functional team to plan and deliver **36 technical sessions** across the academic year, reaching over **200 students** – covering curriculum design, stakeholder communication, and team leadership.

Junior Teaching Assistant | GUC, Introduction to Programming (CSEN 102)

Sep 2023 – Jan 2024

- Ran weekly Python lab sessions for **20+ students** covering data structures, algorithms, and object-oriented programming fundamentals.

CERTIFICATIONS

Data Structures and Algorithms (Microsoft): Algorithmic problem solving and optimization. *(November 2025)*

Web Development (Microsoft and Sprints): Python OOP, RESTful API design, HTML5, CSS3. *(October 2025)*

Machine Learning and AI (Microsoft and Sprints): End-to-end ML pipelines, exploratory data analysis (EDA), Pandas, Scikit-learn, and model deployment. *(September 2025)*

DevOps (Microsoft and Sprints): SDLC, Docker, CI/CD, and AWS (EC2, S3, RDS). *(August 2025)*

Google Cloud Fundamentals: Core Infrastructure (Google Cloud): Core GCP services, IAM, networking, and resource hierarchy.

Essential Google Cloud Infrastructure: Foundation (Google Cloud): Compute Engine, storage services, and deployment basics.

Essential Google Cloud Infrastructure: Core Services (Google Cloud): Load balancing, autoscaling, monitoring, and cloud operations.

Getting Started with Google Kubernetes Engine (Google Cloud): Cluster architecture, workload deployment, scaling, and managing containerized applications on GKE.

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, TensorFlow (Familiar), Scikit-learn, Computer Vision, NLP, LLMs, LangChain, Agentic AI, RAG Systems, QLoRA Fine-Tuning, Prompt Engineering, Model Training & Optimization, Hyperparameter Tuning (Optuna), Vision Transformers (ViT), Explainable AI (SHAP, LIME)

Data Engineering: Apache Spark (PySpark), Apache Kafka, Apache Airflow, ETL Pipelines, Stream Processing, Data Warehousing, Apache Superset, Data Pipeline Orchestration

Databases & Graph: PostgreSQL, MySQL, MongoDB, MSSQL, Redis, Neo4j, SQL, Database Design, Query Optimization, Knowledge Graphs, Vector Databases

Backend & APIs: Java, Spring Boot, Node.js, Express.js, TypeScript, REST APIs, GraphQL, OAuth 2.0, Microservices, Event-Driven Architecture, Message Queues (Kafka, RabbitMQ), Webhooks, WSO2, IBM DataPower

Frontend: React, Remix, Next.js, JavaScript, HTML5, CSS3, Responsive Design, Accessibility (WCAG)

Cloud & DevOps: Docker, Kubernetes, Git, GitHub, Linux, GCP, AWS (EC2, S3, RDS), CI/CD Pipelines, Prometheus, Grafana

Testing & Quality Assurance: Unit Testing, Integration Testing, Vitest, Supertest, Postman, Debugging, Regression Testing

Programming Languages: Java, Python, JavaScript, TypeScript, C, C++, SQL, HTML, CSS

Methodologies & Concepts: Agile/Scrum, SDLC, System Design, Distributed Systems, Object-Oriented Design (OOP), Design Patterns, Data Structures and Algorithms, Code Review, Cross-Functional Collaboration, Networking, Version Control